Program Listing

ELL782 Assignment 2 · Dharun Thota (Entry No. 2026EET2401)

Program

`program.asm` (source) and `program.hex` (Logisim ROM image, `v2.0 raw`, one 16-bit word per line, address 0x00 first). 154 words at ROM addresses 0x00–0x99; HALT at 0x99; 16-bit sum of all words 0x60EB. Unused ROM is 0x0000 (NOP).

Mode parameters

Copied to RAM 0x00–0x04 when START is accepted; MODE is sampled once, together with START.

Parameter	RAM	LIGHT (MODE = 00)	HEAVY (MODE = 01, 10, 11)
Fill target: fill until WATER_LEVEL ≥	0x00	100 (0x64)	180 (0xB4)
Wash half-period (timer ticks per direction)	0x01	10	16
Wash cycles (forward + reverse pairs)	0x02	2	3
Spin time (timer ticks)	0x03	12	24
Drain threshold: drain until WATER_LEVEL ≤	0x04	8	8

DISPLAY codes: 0x01 idle, 0x02 fill, 0x03 wash, 0x04 drain, 0x05 spin, 0x0D done.

Assembler symbols

Symbol	Value	Meaning
`IN_STATUS`	`0xF0`	bit0 START, bit1 DOOR_CLOSED, bits3:2 MODE, bit4 START_PEND, bit5 TRIP
`WATER_LEVEL`	`0xF1`	
`DRUM_RPM`	`0xF3`	
`ACT`	`0xF4`	bit0 VALVE_IN, 1 HEATER, 2 MOTOR_EN, 3 MOTOR_DIR, 4 PUMP_OUT, 5 DOOR_LOCK, 6 BUZZER
`DISPLAY`	`0xF5`	
`SYS_CTRL`	`0xF6`	write bit0 = ACK_START
`T_LIMIT`	`0xF9`	
`T_CTRL`	`0xFA`	write bit0 = EN, bit1 = CLR
`T_STATUS`	`0xFB`	bit0 MATCH, bit1 EN

Listing

Addr	Hex	Label	Assembly	Comment
0x00	9800	start:	LDI r0, 0	
0x01	9FF4		LDI r7, ACT	
0x02	3007		ST r0, [r7]	all actuator requests off
0x03	9FFA		LDI r7, T_CTRL	
0x04	3007		ST r0, [r7]	timer disabled
0x05	9FF5		LDI r7, DISPLAY	
0x06	9901		LDI r1, 0x01	
0x07	3107		ST r1, [r7]	DISPLAY 01 = idle
0x08	9FF0	wait_start:	LDI r7, IN_STATUS	
0x09	B307		LD r3, [r7]	r3 = status byte (keeps the MODE snapshot)
0x0A	A103		MOV r1, r3	
0x0B	9A03		LDI r2, 0x03	START | DOOR_CLOSED
0x0C	E102		AND r1, r2	
0x0D	D102		SUB r1, r2	0 only if both bits are 1
0x0E	5908		BNEZ r1, wait_start	
0x0F	9FF6		LDI r7, SYS_CTRL	
0x10	9901		LDI r1, 0x01	
0x11	3107		ST r1, [r7]	ACK_START
0x12	9FF4		LDI r7, ACT	
0x13	9920		LDI r1, 0x20	
0x14	3107		ST r1, [r7]	DOOR_LOCK on
0x15	9FF9		LDI r7, T_LIMIT	
0x16	9902		LDI r1, 0x02	
0x17	3107		ST r1, [r7]	lock settle time = 2 ticks
0x18	9FFA		LDI r7, T_CTRL	
0x19	9903		LDI r1, 0x03	
0x1A	3107		ST r1, [r7]	timer clear + enable
0x1B	9FFB	settle:	LDI r7, T_STATUS	
0x1C	B107		LD r1, [r7]	
0x1D	9A01		LDI r2, 0x01	
0x1E	E102		AND r1, r2	
0x1F	491B		BEQZ r1, settle	wait for MATCH
0x20	9FF0		LDI r7, IN_STATUS	
0x21	B107		LD r1, [r7]	
0x22	9A02		LDI r2, 0x02	DOOR_CLOSED
0x23	E102		AND r1, r2	
0x24	5929		BNEZ r1, door_ok	
0x25	9800		LDI r0, 0	door opened during locking: unlock and start over
0x26	9FF4		LDI r7, ACT	
0x27	3007		ST r0, [r7]	
0x28	6800		JMP start	
0x29	9A0C	door_ok:	LDI r2, 0x0C	
0x2A	E302		AND r3, r2	r3 = MODE << 2

Addr	Hex	Label	Assembly	Comment
0x2B	5B31		BNEZ r3, heavy	
0x2C	9964	light:	LDI r1, 100	fill target
0x2D	9A0A		LDI r2, 10	wash half-period (ticks)
0x2E	9B02		LDI r3, 2	wash cycles
0x2F	9C0C		LDI r4, 12	spin time (ticks)
0x30	6835		JMP store	
0x31	99B4	heavy:	LDI r1, 180	
0x32	9A10		LDI r2, 16	
0x33	9B03		LDI r3, 3	
0x34	9C18		LDI r4, 24	
0x35	9800	store:	LDI r0, 0	
0x36	3100		ST r1, [r0]	RAM[0] fill target
0x37	8801		ADDI r0, 1	
0x38	3200		ST r2, [r0]	RAM[1] wash half-period
0x39	8801		ADDI r0, 1	
0x3A	3300		ST r3, [r0]	RAM[2] wash cycles
0x3B	8801		ADDI r0, 1	
0x3C	3400		ST r4, [r0]	RAM[3] spin time
0x3D	9908		LDI r1, 8	
0x3E	8801		ADDI r0, 1	
0x3F	3100		ST r1, [r0]	RAM[4] drain threshold
0x40	9FF5	fill:	LDI r7, DISPLAY	
0x41	9902		LDI r1, 0x02	
0x42	3107		ST r1, [r7]	DISPLAY 02 = fill
0x43	9FF4		LDI r7, ACT	r7 = ACT from here on
0x44	9921		LDI r1, 0x21	lock | valve
0x45	3107		ST r1, [r7]	
0x46	9800		LDI r0, 0	
0x47	B400		LD r4, [r0]	r4 = target
0x48	9EF1		LDI r6, WATER_LEVEL	
0x49	B106	fill_loop:	LD r1, [r6]	
0x4A	F104		SLTU r1, r4	level < target ?
0x4B	5949		BNEZ r1, fill_loop	
0x4C	9920		LDI r1, 0x20	
0x4D	3107		ST r1, [r7]	valve closed, lock held
0x4E	9EF5	wash:	LDI r6, DISPLAY	
0x4F	9903		LDI r1, 0x03	
0x50	3106		ST r1, [r6]	DISPLAY 03 = wash
0x51	9801		LDI r0, 1	
0x52	B200		LD r2, [r0]	r2 = T
0x53	8801		ADDI r0, 1	
0x54	B300		LD r3, [r0]	r3 = K
0x55	9EF9		LDI r6, T_LIMIT	
0x56	3206		ST r2, [r6]	LIMIT = T
0x57	9EFA		LDI r6, T_CTRL	
0x58	9DFB		LDI r5, T_STATUS	
0x59	9924	wash_loop:	LDI r1, 0x24	
0x5A	3107		ST r1, [r7]	lock | motor, forward

Addr	Hex	Label	Assembly	Comment
0x5B	9903		LDI r1, 0x03	
0x5C	3106		ST r1, [r6]	timer clear + enable
0x5D	B105	w1:	LD r1, [r5]	
0x5E	9801		LDI r0, 0x01	
0x5F	E100		AND r1, r0	
0x60	495D		BEQZ r1, w1	
0x61	992C		LDI r1, 0x2C	
0x62	3107		ST r1, [r7]	lock | motor | reverse
0x63	9903		LDI r1, 0x03	
0x64	3106		ST r1, [r6]	timer clear + enable
0x65	B105	w2:	LD r1, [r5]	
0x66	9801		LDI r0, 0x01	
0x67	E100		AND r1, r0	
0x68	4965		BEQZ r1, w2	
0x69	8BFF		ADDI r3, 0xFF	K = K - 1
0x6A	5B59		BNEZ r3, wash_loop	
0x6B	9920		LDI r1, 0x20	
0x6C	3107		ST r1, [r7]	motor off, lock held
0x6D	9EF5	drain:	LDI r6, DISPLAY	
0x6E	9904		LDI r1, 0x04	
0x6F	3106		ST r1, [r6]	DISPLAY 04 = drain
0x70	9930		LDI r1, 0x30	lock | pump
0x71	3107		ST r1, [r7]	
0x72	9804		LDI r0, 4	
0x73	B400		LD r4, [r0]	r4 = threshold
0x74	9EF1		LDI r6, WATER_LEVEL	
0x75	B106	drain_loop:	LD r1, [r6]	
0x76	A204		MOV r2, r4	
0x77	F201		SLTU r2, r1	threshold < level ?
0x78	5A75		BNEZ r2, drain_loop	
0x79	9EF5	spin:	LDI r6, DISPLAY	
0x7A	9905		LDI r1, 0x05	
0x7B	3106		ST r1, [r6]	DISPLAY 05 = spin
0x7C	9803		LDI r0, 3	
0x7D	B200		LD r2, [r0]	r2 = spin time
0x7E	9EF9		LDI r6, T_LIMIT	
0x7F	3206		ST r2, [r6]	LIMIT = spin time
0x80	9934		LDI r1, 0x34	lock | pump | motor forward
0x81	3107		ST r1, [r7]	
0x82	9EFA		LDI r6, T_CTRL	
0x83	9903		LDI r1, 0x03	
0x84	3106		ST r1, [r6]	timer clear + enable (rearm)
0x85	9DFB		LDI r5, T_STATUS	
0x86	B105	s1:	LD r1, [r5]	
0x87	9801		LDI r0, 0x01	
0x88	E100		AND r1, r0	
0x89	4986		BEQZ r1, s1	
0x8A	9920	done:	LDI r1, 0x20	

Addr	Hex	Label	Assembly	Comment
0x8B	3107		ST r1, [r7]	motor and pump off, lock held
0x8C	9EFA		LDI r6, T_CTRL	
0x8D	9902		LDI r1, 0x02	
0x8E	3106		ST r1, [r6]	timer clear, disabled
0x8F	9EF3		LDI r6, DRUM_RPM	
0x90	B106	rpm_wait:	LD r1, [r6]	
0x91	5990		BNEZ r1, rpm_wait	wait for DRUM_RPM = 0
0x92	9900		LDI r1, 0x00	
0x93	3107		ST r1, [r7]	unlock
0x94	9EF5		LDI r6, DISPLAY	
0x95	990D		LDI r1, 0x0D	
0x96	3106		ST r1, [r6]	DISPLAY 0D = done
0x97	9940		LDI r1, 0x40	
0x98	3107		ST r1, [r7]	BUZZER
0x99	7000		HALT	